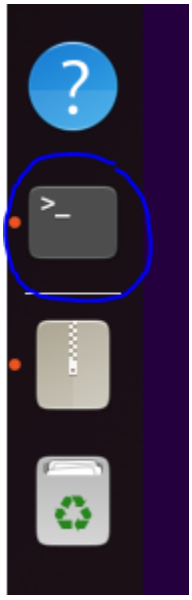
Radius instructions

1. Open linux terminal



2. Enter sudo su mode, which allows you to execute commands that require administrative access. Then download freeradius using the apt-get install freeradius command.

```
radius@CCNP-RADIUS:~$ sudo su
root@CCNP-RADIUS:/home/radius# apt-get freeradius
E: Invalid operation freeradius
root@CCNP-RADIUS:/home/radius# apt-get install freeradius
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
freeradius is already the newest version (3.0.26~dfsg~git20220223.1.00ed0241fa-0ubuntu3.2).
The following packages were automatically installed and are no longer required:
  libwpe-1.0-1 libwpebackend-fdo-1.0-1
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 4 not upgraded.
root@CCNP-RADIUS:/home/radius#
```

3. run a "vim" command, on /etc/freeradius/3.0/clients.conf. The command "vim" allows you to directly edit this config file.

```
root@CCNP-RADIUS:/home/radius# vim /etc/freeradius/3.0/clients.conf
```

4. You should see a similar screen to this. If not, scroll up and down using the arrow keys to get to this area.

```
#  i.e. The entry from the smallest possible network.
#
#client private-network-1 {
#       ipaddr          = 192.0.2.0/24
#       secret          = testing123-1
#}

#client private-network-2 {
#       ipaddr          = 198.51.100.0/24
#       secret          = testing123-2
#}

######################################################################
#
#  Per-socket client lists.  The configuration entries are exactly
#  the same as above, but they are nested inside of a section.
#
#  You can have as many per-socket client lists as you have "listen"
#  sections, or you can re-use a list among multiple "listen" sections.
#
#  Un-comment this section, and edit a "listen" section to add:
#  "clients = per_socket_clients".  That IP address/port combination
#  will then accept ONLY the clients listed in this section.
#
#  There are additional considerations when using clients from SQL.
#
#  A client can be link to a virtual server via modules such as SQL.
#  This link is done via the following process:
#
#  If there is no listener in a virtual server, SQL clients are added
#  to the global list for that virtual server.
#
#  If there is a listener, and the first listener does not have a
#  "clients=..." configuration item, SQL clients are added to the
#  global list.
#
#  If there is a listener, and the first one does have a "clients=..."
#  configuration item, SQL clients are added to that list.  The client
#  { ...} ` configured in that list are also added for that listener.
#
#  The only issue is if you have multiple listeners in a virtual
#  server, each with a different client list, then the SQL clients are
#  added only to the first listener.
#
#clients per_socket_clients {
#       client socket_client {
#               ipaddr = 192.0.2.4
#               secret = testing123
#       }
#}
client accesspoint {
        secret = testing123
        ipaddr = 10.0.0.3
}
"/etc/freeradius/3.0/clients.conf" 292L, 8343B
```

5. Press I to enter insert mode which allows to edit the document. Create a client by using the client access point { secret = password, ipaddr = x.x.x.x } command. This will establish your access point as the client of this radius server. A similar command needs to be established on the wireless access point to establish this pc as the server. Enter a password that will remain constant between your AP and the server and the IP address of your wireless access point.

```
client accesspoint {
        secret = testing123
        ipaddr = 10.0.0.3
```

6. press escape to exit insert mode and type in ":wq", which specifies to exit vim mode and save all edits. Keep in mind that typing in ":q" will exit without saving in case of any misconfiguration.

```
:wq
```

7. Next, run a vim, specifying the same directory, but entering the "users" config file.

```
root@CCNP-RADIUS:/home/radius# vim /etc/freeradius/3.0/users
```

8. You should see a screen similar to this. Note the INSERT in all caps at the bottom of the screen, which indicates you are in insert mode.

```
testing Cleartext-Password := "password"
#
#
#       Configuration file for the rlm_files module.
#       Please see rlm_files(5) manpage for more information.
#
#       This file contains authentication security and configuration
#       information for each user.  Accounting requests are NOT processed
#       through this file.  Instead, see 'accounting', in this directory.
#
#       The first field is the user's name and can be up to
#       253 characters in length.  This is followed (on the same line) wit
#       the list of authentication requirements for that user.  This can
#       include password, comm server name, comm server port number, proto
#       type (perhaps set by the "hints" file), and huntgroup name (set by
#       the "huntgroups" file).
#
#       If you are not sure why a particular reply is being sent by the
#       server, then run the server in debugging mode (radiusd -X), and
#       you will see which entries in this file are matched.
#
#       When an authentication request is received from the comm server,
#       these values are tested. Only the first match is used unless the
#       "Fall-Through" variable is set to "Yes".
#
#       A special user named "DEFAULT" matches on all usernames.
#       You can have several DEFAULT entries. All entries are processed
#       in the order they appear in this file. The first entry that
#       matches the login-request will stop processing unless you use
#       the Fall-Through variable.
#
#       Indented (with the tab character) lines following the first
#       line indicate the configuration values to be passed back to
#       the comm server to allow the initiation of a user session.
#       This can include things like the PPP configuration values
#       or the host to log the user onto.
#
#       You can include another `users' file with `$INCLUDE users.other'

#
#       For a list of RADIUS attributes, and links to their definitions,
#       see: http://www.freeradius.org/rfc/attributes.html
#
#       Entries below this point are examples included in the server for
#       educational purposes. They may be deleted from the deployed
#       configuration without impacting the operation of the server.

#


#
# Deny access for a specific user.  Note that this entry MUST
# be before any other 'Auth-Type' attribute which results in the user
# being authenticated.
#
-- INSERT --
```

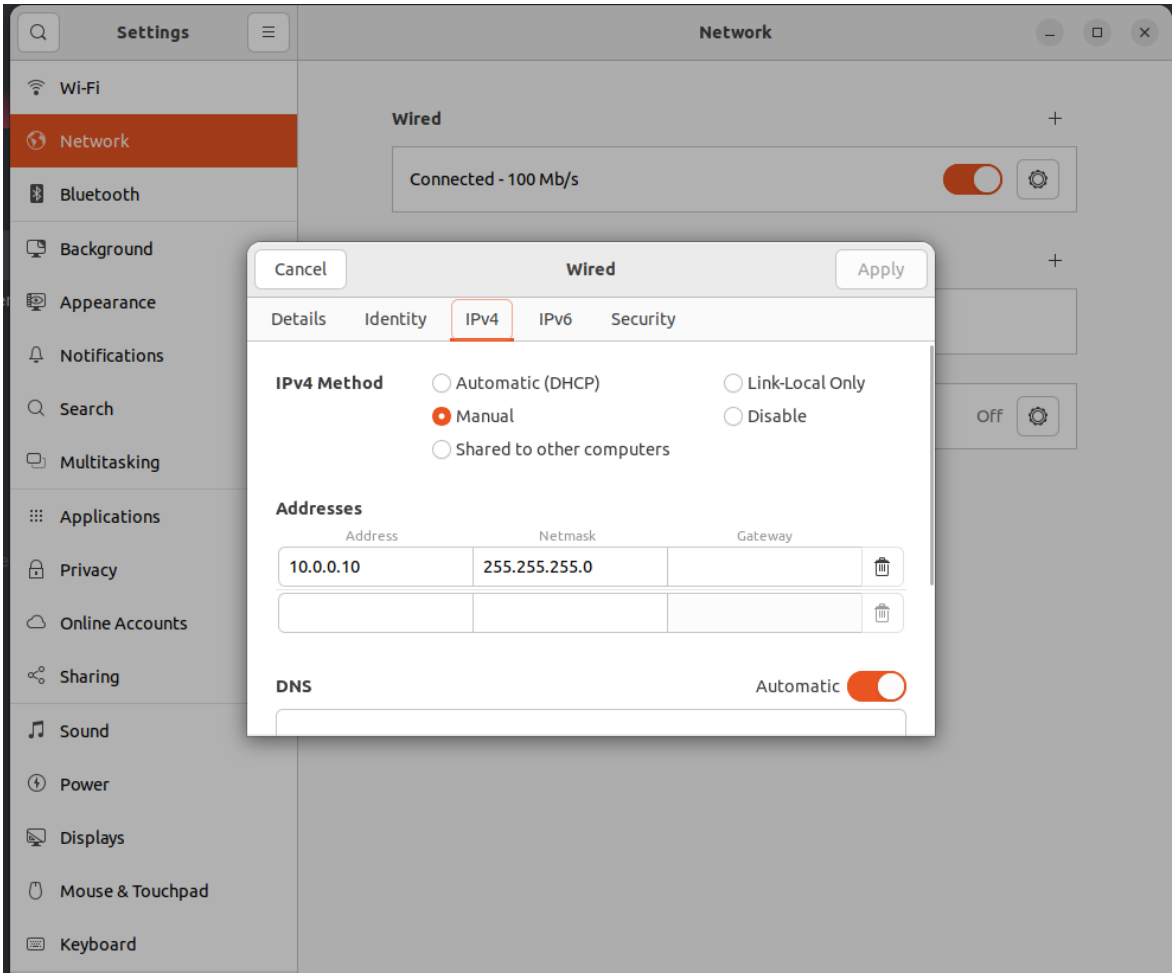
9. Notice the pre-configured user that is already at the top of the file. If there is no pre-configured user, create one by typing (User) Cleartext-Password := "(password)"

```
testing Cleartext-Password := "password"
```

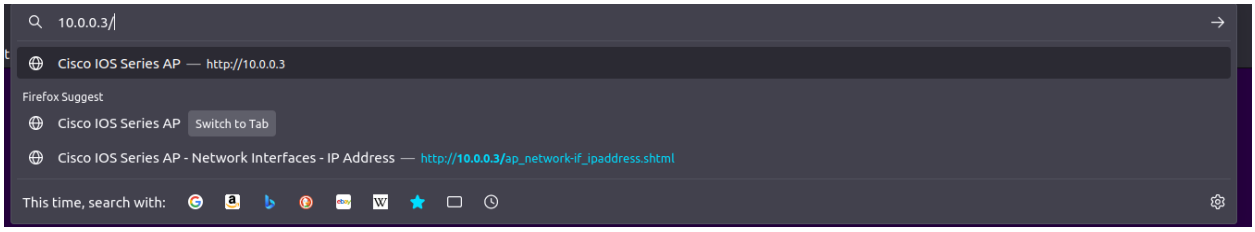
10. Exit insert mode and type in ":wq" again to exit and save.

```
:wq
```

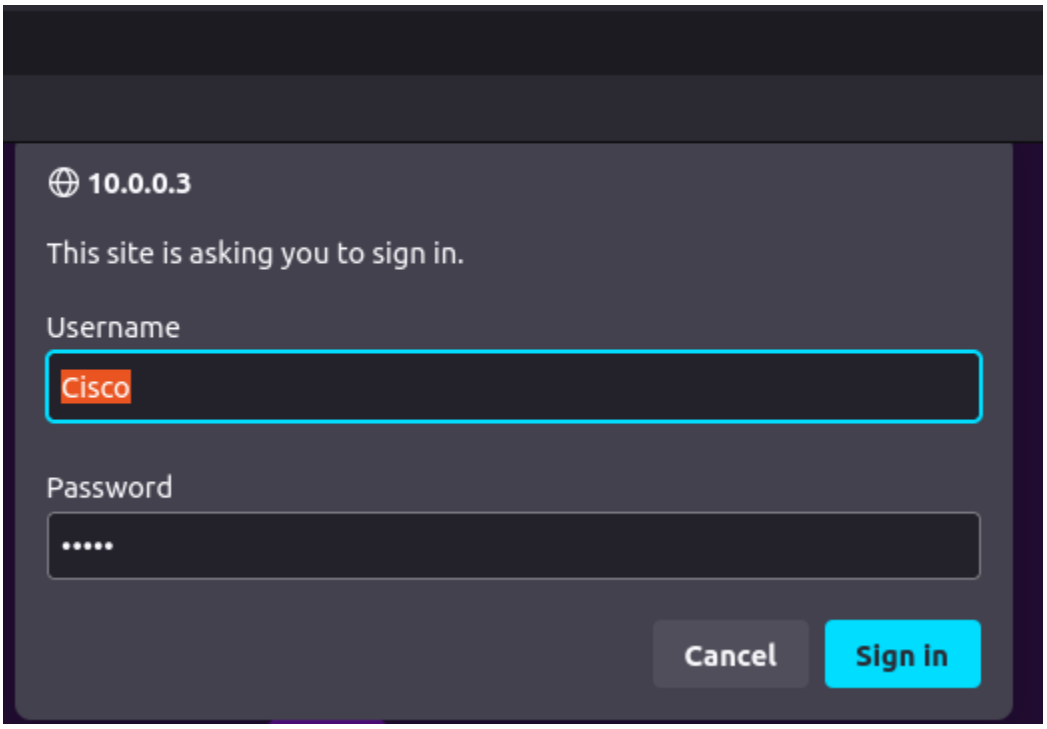
11. Edit your ETHERNET connection and create a static ip address. This IP will be the ip address of your RADIUS server because it is running on your own PC. Remember this IP.



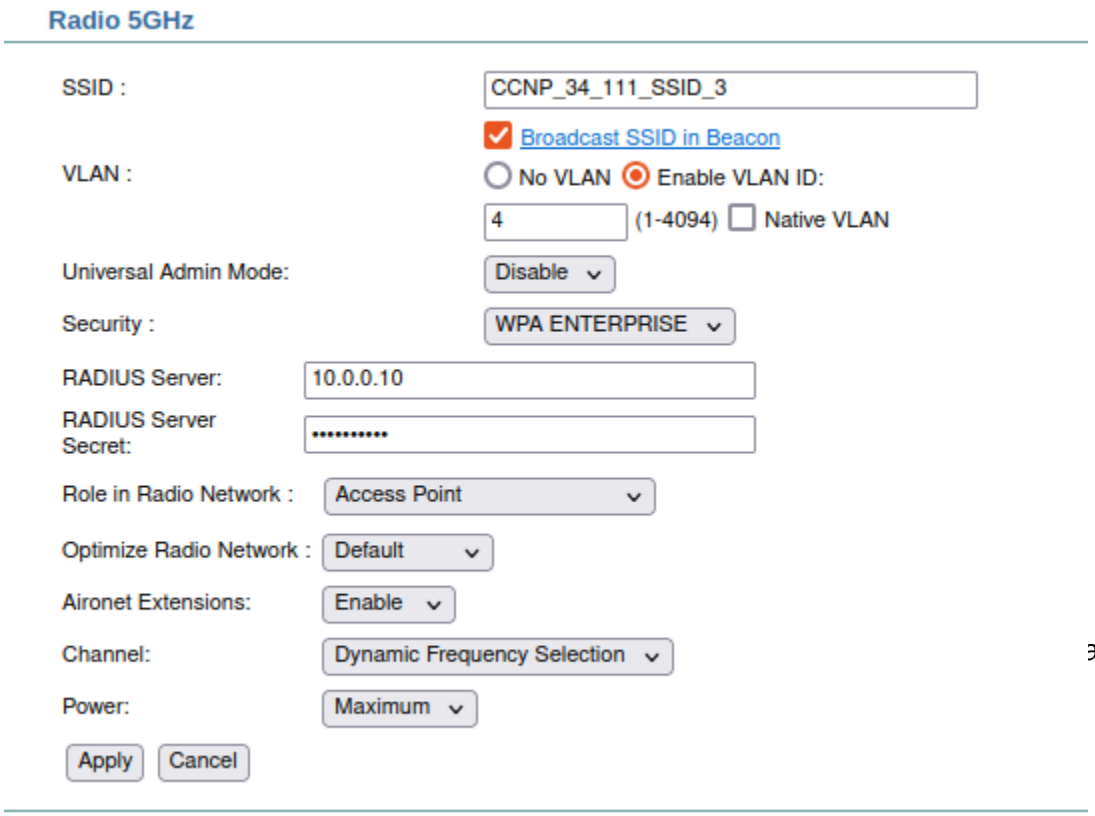
12. enter a new tab and type in the IP address of the wireless AP into the search bar. The default IP is 10.0.0.2 if you have not changed your IP. (though note that if there are multiple of these APs on the same LAN and with the default IP, it will result in overlapping IP addresses).



13. You should get this prompt, type in your username and password and log in. The default username is Cisco and the default password is Cisco



14. Create an SSID and assign it a VLAN. Point it towards the RADIUS server's ip address and type in the RADIUS server's Secret which was configured in step 5.



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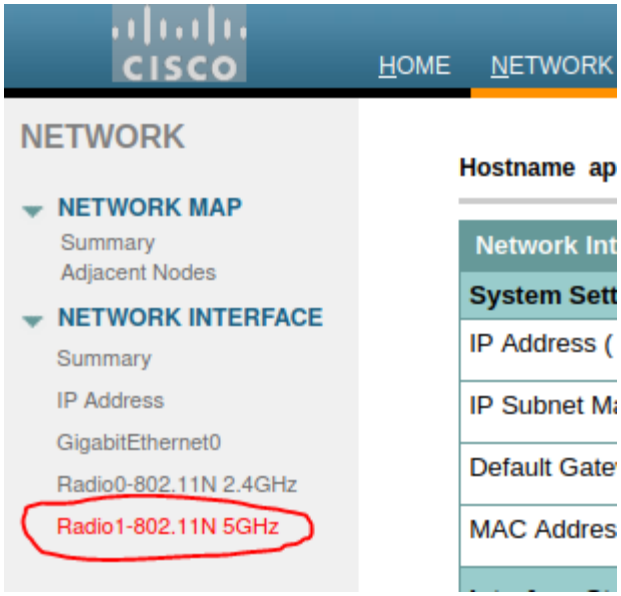
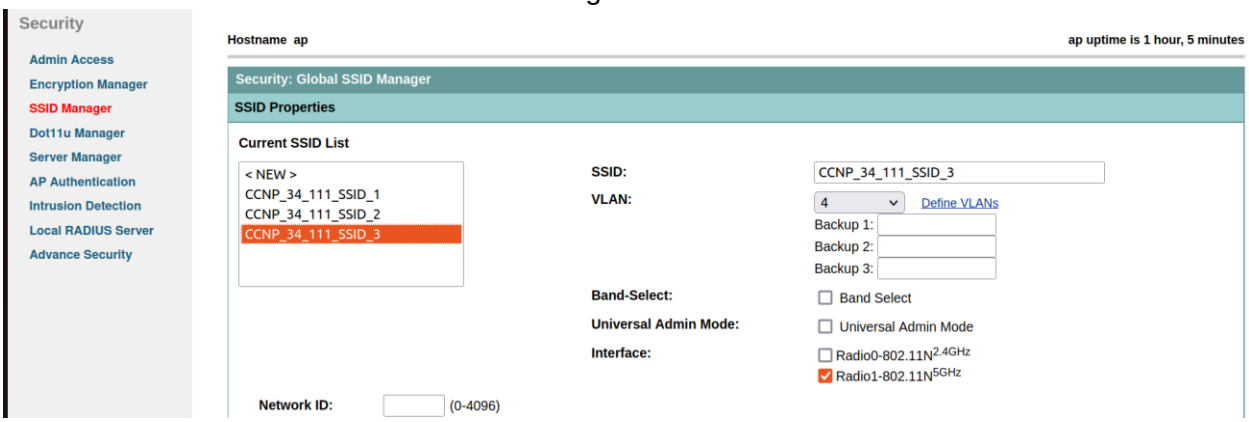
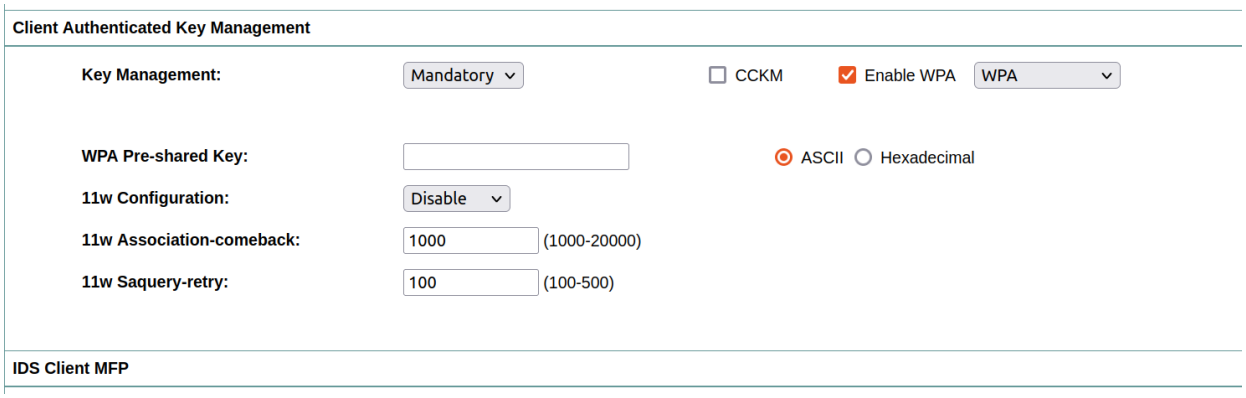


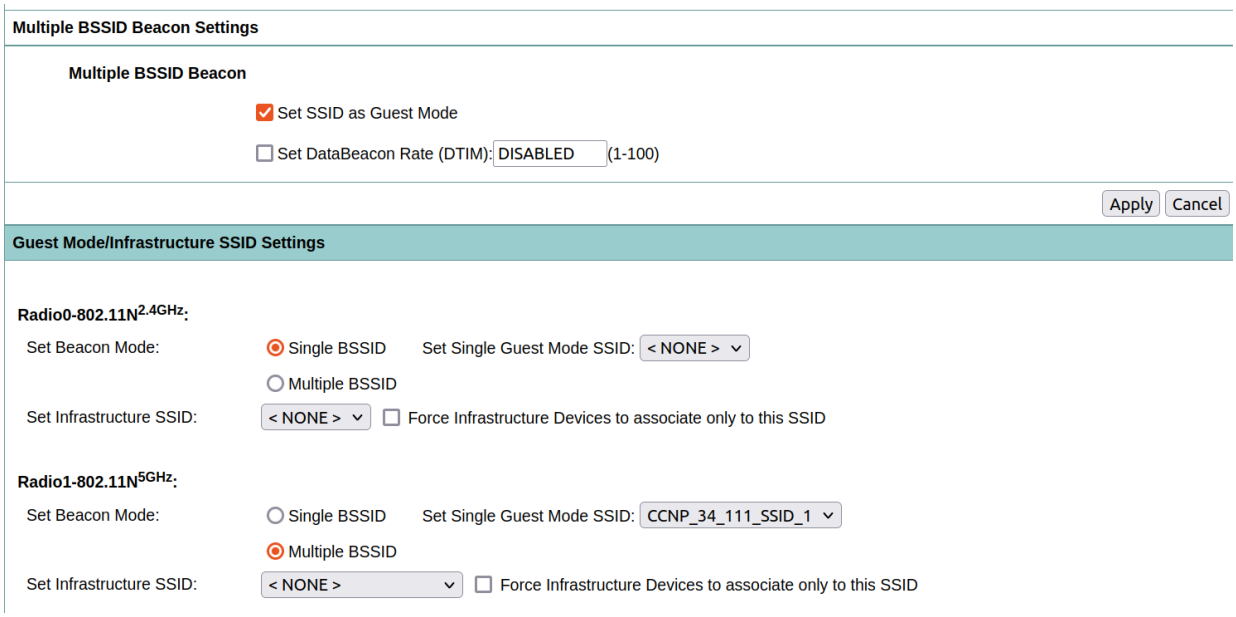
16. Enter Security > SSID Manager and specify that you are configuring the RADIUS SSID. Then make sure that the SSID is using the correct interface and VLAN number.



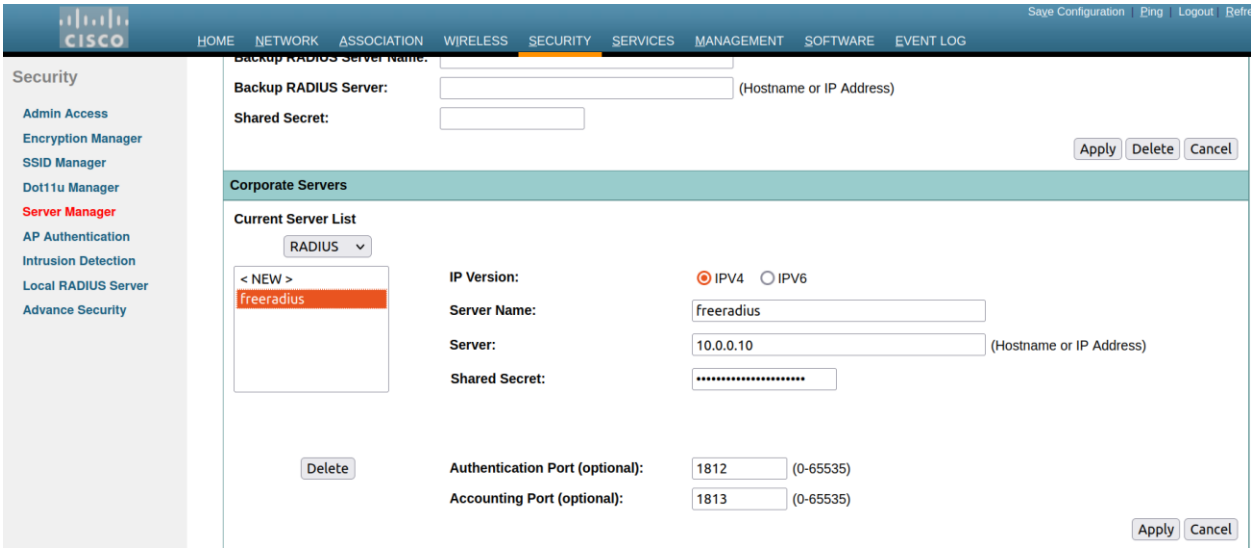
16.5. Scroll down to Client Authentication Key management and allow WPA.



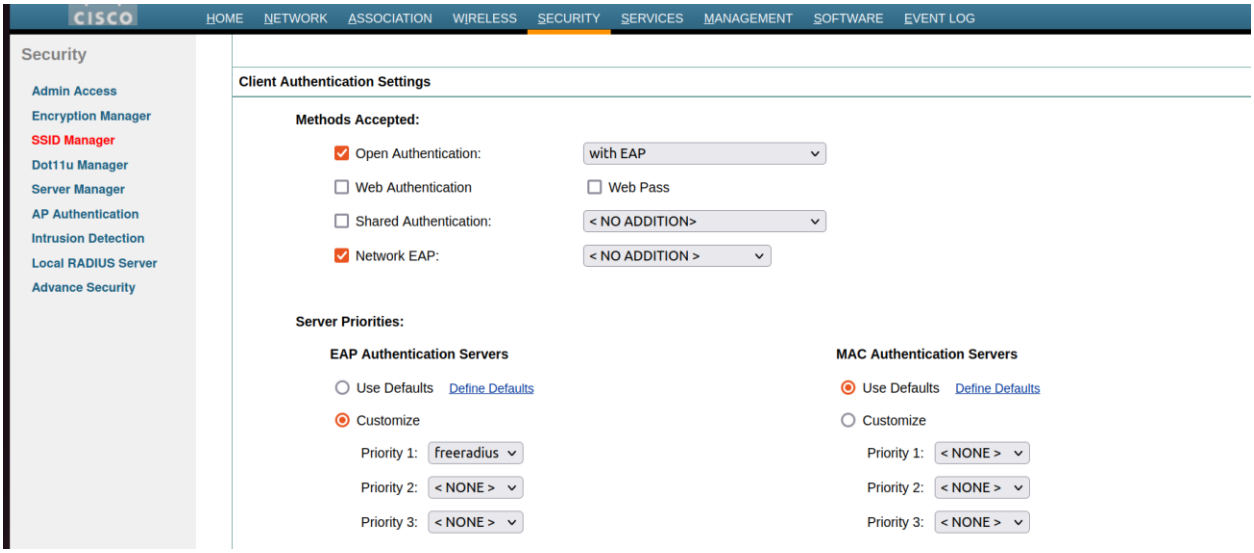
17. Under Multiple BSSID Beacon settings, set the SSID to be in guest mode. Then under Guest/Infrastructure Settings, set the beacon to multiple BSSID mode. This will allow your Beacon to be broadcasted.



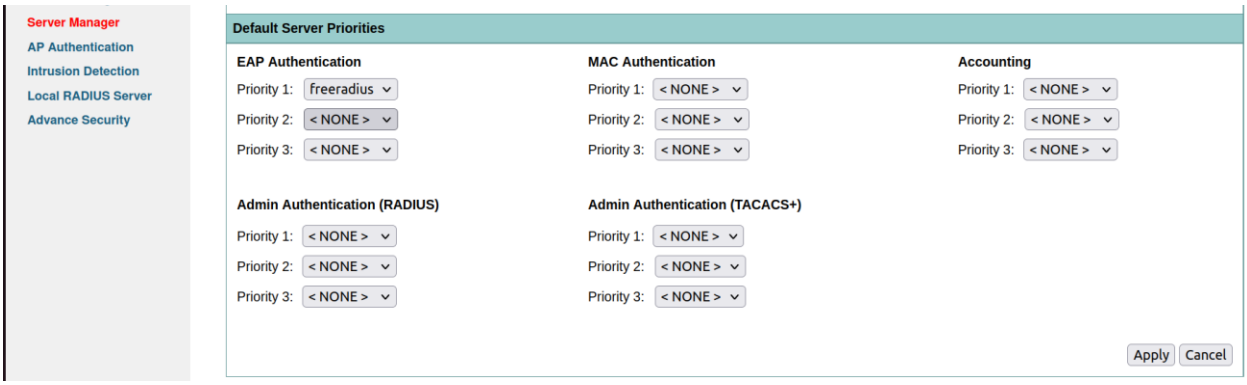
18. Under the Server Managers tab, create a new server, specify the server name, IP, and the shared secret. Specify ports 1812 and 1813 for authentication and accounting.



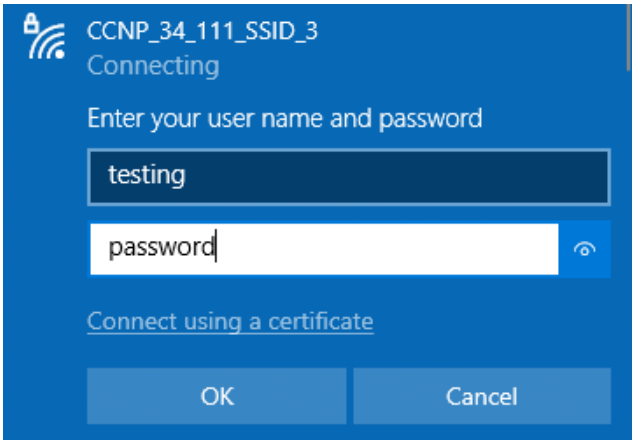
19. Go back into SSID Manager and enable Open Authentication with EAP and Network EAP. Then specify the server priorities to use the radius server you configured in the previous step.



20. After enabling EAP, go back to Server Manager and set priority 1 to use your previously configured radius server.



21. Go back into your radius server's linux config and run a freeradius -X with root privileges.

```
radius@CCNP-RADIUS:~$ sudo freeradius -X
```

22. connect to the RADIUS wifi using the previously configured test user's credentials we created in step 9